

Robust Optimization in Active Linear Regression

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Motivation

Training Cost in Examples

For a fixed compute budget, training a good transformer could require ≈ 20 tokens per parameter, so for 70B parameters we could need 1.4T tokens [Hoffmann et al., 2022].

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Cost of Labelling

The average hourly rate is \$25.23 and experts can be even be paid up-to \$54.09 (40-100k annual)[ZipRecruiter, 2026].

Can we reduce this human cost, through reducing the label complexity?

Linear Regression and Active Learning

ℓ_2 Linear Regression

Given $X \in \mathbb{R}^{N \times d}$ and $y \in \mathbb{R}^N$. Compute $w_* = \arg \min_{w \in \mathbb{R}^d} \|Xw - y\|_2^2$.

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Active Learning

Suppose we are only given X , and entries of y can be queried with **a cost**.

- Obtaining labels is **expensive** and slow, as they require experts (e.g., radiologists, pathologists, clinicians)

Can we reveal less of y while obtaining $\hat{w} \approx w_*$?

What does it mean for $\hat{w} \approx w_*$?

Claim

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Claim

If the algorithm deterministically removes the red rows, we lose all information.

What does it mean for $\hat{w} \approx w_*$?

ϵ -regret

Suppose $\hat{w}$ is drawn from some distribution μ .

$\hat{w}$ approximates w_* with ϵ -regret if $\mathbb{E}_\mu[\|X\hat{w} - y\|_2^2] \leq (1 + \epsilon) \cdot \|Xw_* - y\|_2^2$.

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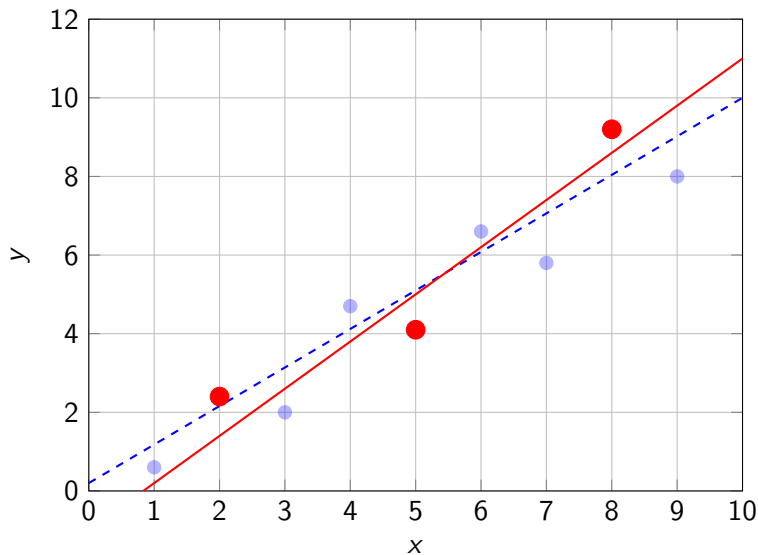
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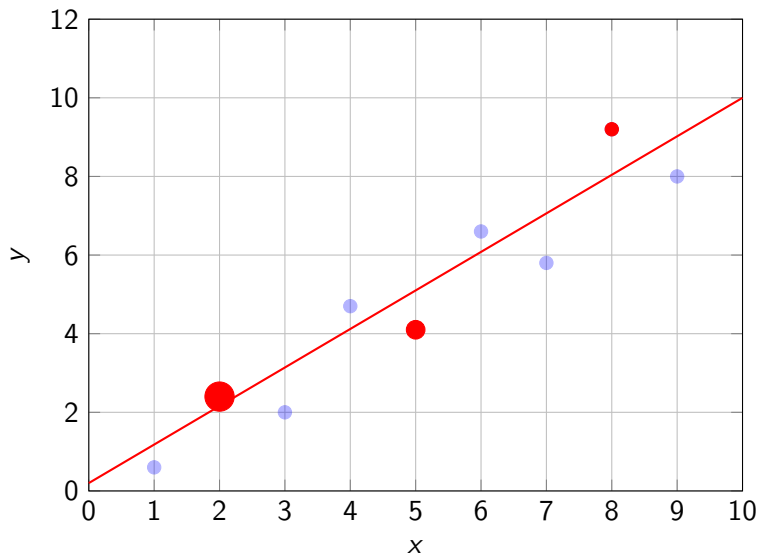
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Coreset

For most known methods, we sample a coreset, i.e., a subset of (rescaled) rows (A, y_A) of (X, y) , and take $\hat{w} = \arg \min_w \|Aw - y_A\|_2^2$.





Why $\epsilon = d/N$?

Main Focus

We will focus on $\epsilon = d/N$ to preserve the benefits of using **all** of X while using **less** of y .

By [Abu-Mostafa et al., 2012][Chapter 3], we know if (X, y) is sampled i.i.d. from a distribution, then

$$\|Xw_* - y\|_2^2 = \left(1 + \mathcal{O}\left(\frac{d}{N}\right)\right) \cdot \text{OPT}.$$

Why $\epsilon = d/N$?

If $\hat{w}$ approximates w_* with regret $\epsilon = d/N$, then $\hat{w}$ generalizes as well as w_* .

Prior Works

Known algorithms are

- Following [Drineas et al., 2008], Chen and Price [2019] proved **leverage score sampling** yields coreset yields $\mathcal{O}(d/\epsilon + d \log d)$ in expectation.
- Chen and Price [2019] proved that [Lee and Sun, 2015]'s **randomized BSS** algorithm can yield a coreset of size $\mathcal{O}(d/\epsilon)$. **This is optimal.** However, the algorithm has a large multiple constant.
- Shimizu et al. [2024] gave a **pivotal dependent sampling algorithm** that yields coreset yields $\mathcal{O}(d/\epsilon + d \log d)$. This has been showed empirically to be better than leverage score sampling.
- [Dereziński and Warmuth, 2017, Dereziński et al., 2018] proposed **volume sampling** and **leverage volume sampling** that yield a coreset of size $\mathcal{O}(d^2/\epsilon)$ and $\mathcal{O}(d/\epsilon + d \log d)$, respectively.

Claim

The exact coreset size and the label saving at $\epsilon = d/N$ are unknown.

Gittens and Magdon-Ismail [2023]

- showed we can remove 1 data point and achieve a regret of $\mathcal{O}(d/N^2)$
- conjectured that removing 1 extra row and only incur an extra $\mathcal{O}(d/N^2)$ -regret (chaining).

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Question

Is this chaining algorithm possible?

Sub-Leverage

We define the sub-leverage as the following:

$$\ell_{A,i} = x_i^\top (X^\top X - A^\top A)^{-1} x_i$$

Theorem (Main Result)

Let A be a set of k rows in X . Sample a row x_i in $X_A^- := X \setminus A$ with $p_{A,i} \propto \frac{(1-\ell_{A,i})^2}{\ell_{A,i}}$. Let $A' = A \cup \{x_i\}$. Then,

$$\mathbb{E} [\|X w_{A'}^- - y\|^2] \leq \|X w_A^- - y\|^2 + \frac{d \cdot (1 + \kappa_A)}{(n - d - k)^2} \|X_A^- w_A^- - y_A^-\|^2 + \mathcal{E}_A$$

where κ_A and $\mathcal{E}_A$ are constants dependent on A (cont.).

Algorithm

Definition

Let $X = U\Sigma V^T$ be the thin SVD of X . Then,

- U_A is the rows of U corresponding to row of A in X and U_A^- is the remaining rows of U .
- Partial projection matrix $P_A = U_A U_A^T$.

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Theorem (Main Result) (cont.)

where $\kappa_A = \|P_A\|_2(2 - \|P_A\|_2)(1 - \|P_A\|_2)^{-2}$ and

$$\mathcal{E}_A = \frac{2}{Z_A} (Aw_A^- - y_A)^T (I_k - P_A)^{-1} U_A (U_A^-)^T D (X_A^- w_A^- - y_A^-)$$

where $Z_A = \sum \frac{(1 - \ell_{A,i})^2}{\ell_{A,i}} \geq (N - d - k)^2 / d$ and

$$D = \text{Diag} \left(\left[\frac{1}{\ell_{A,i}} \right]_i \right) - \frac{1}{N - k} \left(\sum_i \frac{1}{\ell_{A,i}} \right) I_{N-k}$$

Algorithm Assumption

Assumption 1

$\|P_A\|_2 \leq c < 1$ for some c . This implies κ_A is bounded.

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Assumption 2

$\mathbb{E}[\mathcal{E}_A] = 0$. This is appropriate if the residual is Gaussian.

Experimental Setup

Synthetic Setup

$N = 1000$. We consider two datasets using scikit-learn [Pedregosa et al., 2011]:

- A synthetic regression dataset with $d = 10$, and 5 informative features with Gaussian noise with standard deviation 0.002.
- The two moons dataset with $d = 2$ with Gaussian noise with standard deviation 0.05.

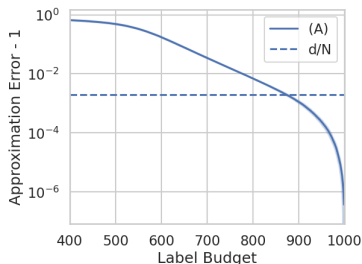
Real Life Datasets

We consider two real world regression datasets:

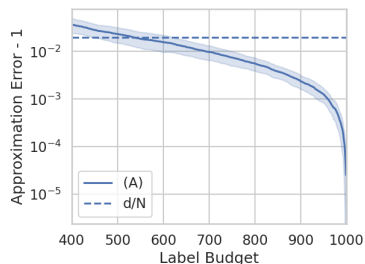
- Wine Quality (red and white wine) [Cortez and Reis, 2009].
- Concrete Compressive Strength [Yeh, 1998].

We remove up-to 60% of the labels in each and measure the amount of budget up-to the approximation ratio that we can remove. No feature engineering is done.

Synthetic Experiments



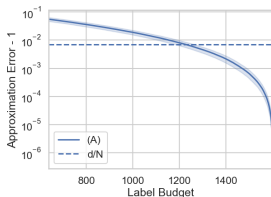
(a) Two moons



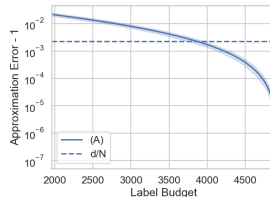
(b) Synthetic Regression

Figure 1: Synthetic datasets. In 1a, we only remove roughly 10% of the rows before we reach the approximation ratio, this is likely due to being a classification problem. In 1b, we can remove nearly 50% of the labels in the dataset matching our conjecture.

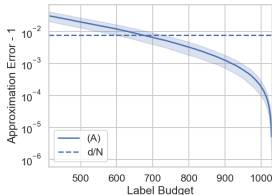
Real World Experiments



(a) Red Wine Quality



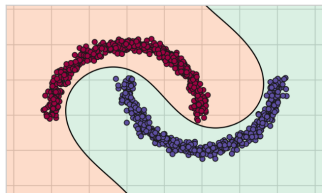
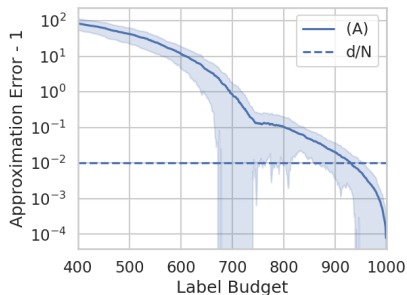
(b) White Wine Quality



(c) Concrete Strength

Figure 2: We remove roughly 30% of the labels before our approximation ratio is achieved in the wine datasets. The concrete dataset removes roughly 50%.

Limitations & Future Works



- Fitting a better regression model for the moons dataset (3rd degree polynomial) does not yield better label budget.
- Some synthetic datasets do not meet the necessary assumptions of this method to get a good label budget. Constants are still unknown.
- Developing more relaxed assumptions would be a good future work. Additional work could be done towards tightening the approximation bound's relationship to the label budget.

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Thank You!